

Tab 1

CroudSourced Civic Issue Reporting and Resolution System

Problem Background

Every day, citizens face problems like *potholes*, *broken streetlights*, *garbage bins overflowing*, and *waterlogging*. But complaints are ignored or lost, with no transparency or incentives for citizens. This creates frustration and delays in fixing civic issues.

Goal

To build a transparent, fast, and gamified(reward based) platform where citizens can report issues easily, the government can track and resolve them, and progress is visible to everyone while citizens get rewards for participation.

Proposed Solution – Civic Connect

A Web + Mobile PWA platform connecting citizens and authorities. Citizens report issues with photo + location, track status, earn rewards, and escalate if not resolved. Govt. Officials use a dashboard to assign, monitor, and analyze complaints.

Key Features

1. Photo & location-based reporting
2. Gamification & rewards
3. Time-bound resolution (SLA)
4. Escalation mechanism
5. Crowd verification
6. AI-powered validation
7. Analytics & transparency dashboards

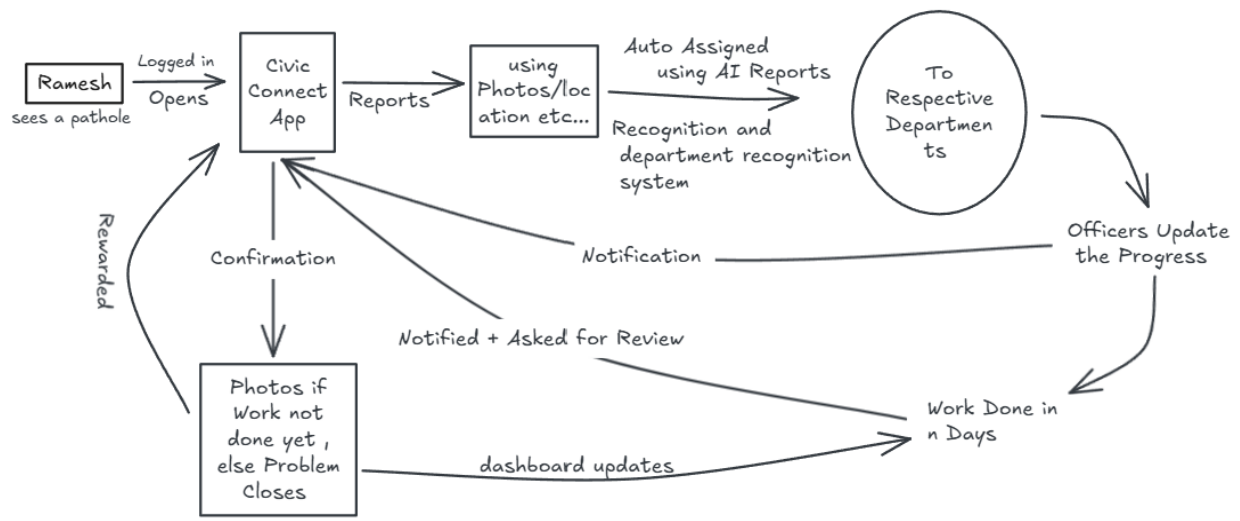
Tech Stack

Frontend: React + Tailwind + PWA
Backend: Node.js + Express
Database: MongoDB / PostgreSQL
Auth: Clerk / Firebase Auth
AI Layer: TensorFlow.js / OpenCV
Storage: AWS S3 / Firebase Storage
Maps: Google Maps / Mapbox
Notifications: Firebase Cloud Messaging

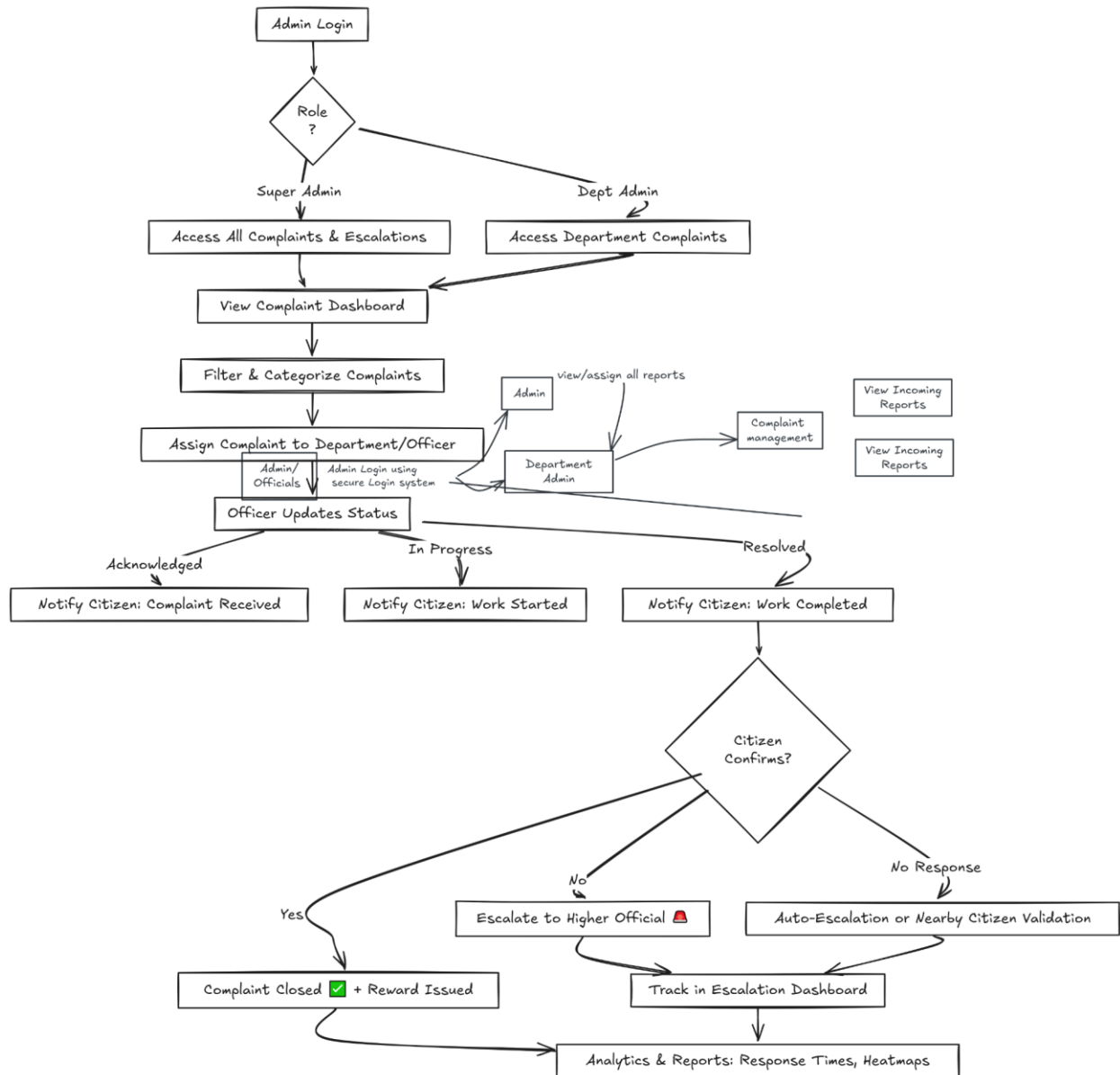
Workflow

1. Citizen submits complaint with photo + location
2. System stores report and auto-assigns to dept.
3. Official receives and updates status
4. Resolution updated → marked Done
5. User verification (Done/Not Done with proof)
6. Crowd verification or auto-close if user inactive
7. Rewards distributed and escalation if deadlines missed

User Journey Example



Admin Work Flow



SDLC Approach

1. Requirement analysis
2. Planning & team roles
3. Design (wireframes, ER diagrams)
4. Development (app + dashboard)
5. Testing (functional, usability, security)
6. Deployment (cloud hosting)
7. Feedback & iteration

Risks & Mitigation

Fake reports → AI + crowd verification

User inactive → auto-close + crowd confirmation

False closure by dept. → citizen proof + escalation

Low adoption → gamification + govt. Integration

Benefits

For Citizens: Faster resolution, rewards, transparency

For Government: Efficiency, accountability, analytics

For Society: Cleaner, safer, connected cities

Future Scope

1. AI-powered insights & predictions
2. Multi-language & accessibility support
3. Integration with Smart Cities, Swachh Bharat, municipal portals
4. Blockchain for transparency & anti-corruption
5. Expansion to road safety, women safety, disaster alerts
6. Reward ecosystem via partnerships with businesses & govt. Services

Value Proposition

Citizens: Easy reporting + rewards

Government: Accountability & efficiency

Businesses: CSR visibility & partnerships

Why Our System is Better than Current Solutions in India?

1. *Limitations of Current Systems :*

- Existing apps (e.g., Swachhata App) focus mainly on reporting but often lack follow-up.
- Reports are sometimes marked resolved without real verification.
- Little to no reward/recognition for citizens.
- Escalation to higher officials is missing or slow.

2. *Our Improvements :*

- **Gamification & Rewards:** Citizens earn points, badges, or redeemable rewards for valid complaints.
- **Two-Step Verification:** Work completion is validated by the reporting user and optionally nearby citizens.
- **Escalation:** Unresolved issues escalate automatically to higher authorities.
- **Transparency Dashboard:** Citizens and officials can track real-time progress.
- **AI Insights:** Cluster issues by type and location to identify priority zones.
- **Mobile-First PWA:** Works on low-end devices without needing heavy app installation.

How It Ensures Work is Done (Beyond Just Reporting)

1. **Mandatory Verification:** A complaint is not closed until validated by the user or nearby citizens.
2. **Escalation Workflow:** If not verified in time, it escalates to higher authorities.
3. **Accountability Metrics:** Officer performance is tracked by resolution time and citizen ratings.
4. **Citizen Empowerment:** Users can challenge false or unresolved closures.

5. ***Useful Rewards***: Redeemable in UPI vouchers, metro recharge, FASTag, or government services.

Conclusion

In short, while most systems stop at reporting, our solution ensures real completion, citizen rewards, transparent accountability, and AI-driven governance improvements.